1.3 PHYSIOGRAPHY AND AGRO-ECOLOGICAL ZONES (AEZ)

The land use pattern of the country is influenced by agro ecology, soil physiographic and climatic factors. According to the variations of all these factors and agricultural potential, the total land area has been classified into thirty agro ecological zones which are grouped into twenty major physiographic units as shown below.

Sl. No	Physiographic Unit	Agro-Ecological Zone (AEZ)		Area excluding river beds (acre)	Texture = T pH = pH Organic Matter Status = 0.m	Land Type					
		AEZ No.	Name			High	Medium High	Medium Low	Low	Very Low	Home-stead + Water
1	2	3	4	5	6	7	8	9	10	11	12
1.	Old Himalayan Piedmont Plain	1.	Old Himalayan Piedmont Plain	990400	Type (%) T : pH: 0.m:	58 Sandy loam 4.5-5.5 Low	34 Loamy 4.5-5.5 Low	1 - - -	- - - -	- - - -	7 - - -
2.	Teesta Flood Plain	2.	Active Teesta Flood Plain	205100	Type (%) T : pH: 0.m:	2 - - -	72 Sandy loam 6.0-6.6 Low	- - - -	- - - -	- - - -	26 - - -
		3.	Teesta Meander Flood Plain	2339595	Type (%) T : pH: 0.m:	35 Loamy 5.4-6.5 Low	51 Loamy 6.0-6.5 Low	4 - - -	1 - - -	- - - -	9 - - -
3.	Teesta Flood Plain	4.	Karatoya-Bangali Flood Plain	635555	Type (%) T : pH: 0.m:	23 Silt-loam 5.4-5.7 Low	44 Clayey 5.4-5.7 Low	14 - - -	4 - - -	1 - - -	14 - - -
4.	LowerAtraiBasin	5.	LowerAtraiBasin	210285	Type (%) T : pH: 0.m:	2 - - -	8 - - -	21 Clayey 4.8-6.0 Medium	65 Clayey 4.8-6.0 Medium	- - - -	4 - - -
5.	Lower Punarbhaba Flood Plain	6.	Lower Punarbhaba Flood Plain	31875	Type (%) T : pH: 0.m:	- - - -	10 - - -	60 Clayey 4.5 Medium	- - - -	- - - -	30 - - -
6.	Brahmaputra Flood Plain	7.	Active Brahmaputra-Jamuna Flood Plain	788265	Type (%) T : pH: 0.m:	5 - - -	37 Sandy/Silty 7.5-7.9 Very low	20 Silty 7.5-7.9 Low	8 - - -	- - - -	30 - - -
		8.	Young Brahmaputra-Jamuna Flood Plain	1463850	Type (%) T : pH: 0.m:	18 Loamy 5.5-6.8 Low	42 Loamy 5.5-6.8 Low	19 Loamy 5.5-6.8 Low	9 - - -	- - - -	12 - - -
		9.	Old Brahmaputra Flood Plain	1786570	Type (%) T : pH: 0.m:	28 Silt-loam 5.1-5.6 Low	35 Loamy 5.1-5.6 Medium	20 Clayey 5.1-5.6 Medium	7 - - -	- - - -	10 - - -

Sl. No	Physiographic Unit	Agro-Ecological Zone (AEZ)		Area excluding river beds (acre)	Texture = T pH = pH Organic Matter Status = 0.m	Land Type					
		AEZ No.	Name			High	Medium High	Medium Low	Low	Very Low	Home-stead + Water
1	2	3	4	5	6	7	8	9	10	11	12
7.	GangesRiver Flood Plain	10.	Active Ganges Flood Plain	823850	Type (%) T : pH: 0.m:	12 - - -	33 Loamy 6.9-7.9 Low	18 Clayey 6.9-7.9 Medium	4 - - -	- - - -	33 - - -
		11.	HighGanges River Flood Plain	3263025	Type (%) T : pH: 0.m:	43 Silt-loam 6.1-7.9 Low	32 Loamy 6.1-7.9 Low	12 Clayey 6.1-7.9 Medium	2 - - -	- - - -	11 - - -
		12.	LowGangesRiver Flood Plain	1968935	Type (%) T : pH: 0.m:	13 Silt-loam 6.2-7.7 Medium	29 Silt-clay 6.2-7.7 Medium	31 Clayey 6.2-7.4 Medium	14 - - -	2 - - -	11 - - -
8.	Ganges Tidal Flood Plain	13.	Ganges Tidal Flood Plain	4217100	Type (%) T : pH: 0.m:	2 - - -	78 Loamy 6.5-7.0 Medium	2 - - -	- - - -	- - - -	18 - - -
9.	Gopalganj-Khulna Beel	14.	Gopalganj-Khulna Bils	555245	Type (%) T : pH: 0.m:	3 - - -	13 Clayey 5.4 Medium	41 Clayey 5.4 Medium	28 Clayey 5.4 Medium	11 - - -	4 - - -
10.	Arial Beel	15.	Arial Beel	35585	Type (%) T : pH: 0.m:	- - - -	13 Clayey 5.4 Medium	- - - -	73 Clayey 5.4 Medium	- - - -	14 - - -
11.	MeghnaRiver Flood Plain	16.	MiddleMeghnaRiver Flood Plain	384250	Type (%) T : pH: 0.m:	- - - -	8 - - -	29 Loamy 4.9-5.5 Low	25 Loamy 4.4-5.5 Low	11 - - -	27 - - -
		17.	LowerMeghnaRiver Flood Plain	224620	Type (%) T : pH: 0.m:	14 Silt-loam 5.0-6.0 Medium	28 Silt-loam 5.0-6.0 Medium	31 Silt-loam 5.5-6.0 Medium	- - - -	- - - -	27 - - -
12.	Meghna Estuarine Flood Plain	18.	Young Meghna Estuarine Flood Plain	2290420	Type (%) T : pH: 0.m:	- - - -	45 Silt-loam 6.1-6.8 Medium	7 - - -	- - - -	- - - -	48 - - -
		19.	Old Meghna Estuarine Flood Plain	1912595	Type (%) T : pH: 0.m:	2 - - -	24 Silt-loam 5.0-6.1 Medium	33 Loamy 5.5-6.5 Medium	21 Clayey 5.5-6.5 Medium	3 - - -	17 - - -
13.	Surma-Kusiyara Flood Plain	20.	Eastern Surma-Kusiyara Flood Plain	1142120	Type (%) T : pH: 0.m:	5 - - 	25 Silt-loam 4.7-6.9 Medium	20 Clayey 4.7-5.5 Medium	36 Clayey 4.7-5.5 Medium	- - - -	14 - - -
		21.	SylhetBasin	1130015	Type (%) T : pH: 0.m:	- - - -	4 - - -	19 Silt-loam 4.7-4.9 High	43 Clayey 4.7-4.9 High	23 Clayey 4.7-4.9 High	11 - - -
14.	Northern and Eastern Piedmont Plains	22.	Northern and Eastern Piedmont Plains	997810	Type (%) T : pH: 0.m:	33 Sandy-loam 4.5-5.8 Medium	31 Loamy 4.5-5.8 Medium	16 Silt-loam 4.5-5.8 Medium	9 - - -	1 - - -	10 - - -

Sl. No	Physiographic Unit	Agro-Ecological Zone (AEZ)		Area excluding river beds (acre)	Texture = T pH = pH Organic Matter Status = 0.m	Land Type					
		AEZ No.	Name			High	Medium High	Medium Low	Low	Very Low	Home-stead + Water
1	2	3	4	5	6	7	8	9	10	11	12
15.	Chittagong Coastal Plain	23.	Chittagong Coastal Plain	919230	Type (%) T : pH: 0.m:	17 Silt-loam 5.6 Low	43 Silt-loam 5.6 Low	13 - - -	- - - -	- - - -	27 - - -
16.	St. Martins Coral Island	24.	St. Martins Coral Island	1975	Type (%) T : pH: 0.m:	33 Sandy 7.0-7.5 Low	63 Sandy-loam 7.0-7.5 Low	2 - - -	- - - -	- - - -	2 - - -
17.	Barind Tract	25.	Level Barind Tract	1247635	Type (%) T : pH: 0.m:	30 Loamy 5.0-5.7 Low	55 Loamy 5.0-5.7 Low	4 - - -	2 - - -	- - - -	9 - - -
		26.	High Barind Tract	395370	Type (%) T : pH: 0.m:	93 Loamy 4.8-5.9 Low	1 - - -	- - - -	- - - -	- - - -	6 - - -
		27.	North-Eastern Barind Tract	266625	Type (%) T : pH: 0.m:	36 Loamy 4.8-5.9 Low	56 Loamy 4.8-5.9 Low	1 - - -	- - - -	- - - -	7 - - -
18.	Madhupur Tract	28.	Madhupur Tract	1048715	Type (%) T : pH: 0.m:	56 Loamy 4.8-5.5 Low	18 Loamy 4.8-5.5 Low	7 - - -	9 - - -	- - - -	10 - - -
19.	Northern and Eastern Hills	29.	Northern and Eastern Hills	4490150	Type (%) T : pH: 0.m:	92 Loamy 4.5-4.9 Low	2 - - -	1 - - -	- - - -	- - - -	5 - - -
20.	Akhaura Terrace	30.	Akhaura Terrwsace	27920	Type (%) T : pH: 0.m:	55 Loamy 5.5-6.5 Low	11 Loamy 5.5-6.5 Low	10 - - -	15 - - -	3 - - -	6 - - -

Source: (i) Land Resources Appraisal of Bangladesh for Agricultural Development, Technical Report No. 2 UNDP/FAO.
(ii) BARC Soils Publication No. 32.

1.4 Agro-Ecological Zones (AEZ) :

AEZ-1: Old Himalayan Piedmont Plain

This distinctive region is developed in an old Teesta alluvial fan extending out from foot of the Himalayas. It has complex relief pattern comprising broad and narrow floodplain ridges and linear depressions. Deep rapidly permeable sandy loams and sandy clay loams are predominant in the region. They are strongly acidic in topsoil and moderately acidic in subsoil as well as richin sandminerals. Seven general soil types occur in the region; of them, Noncalcareous Brown Floodplain soils, Black Terai soils and Noncalcareous Dark Grey Floodplain soils predominent.Organic matter contents are relatively higher than other floodplain areas. The natural fertility of the soils, except the coarse textured, is moderate but well sustained.

Location: Most of Panchagarh and Thakurgaon districts and north western part of Dinajpur district

AEZ-2: Active Tista Floodplain

This region includes the active floodplains of Teesta, Dharla and Dudkumar rivers. It has complex patterns of low, generally smooth ridges, inter-ridge depressions, river channels and cut-off channels.

The area has irregular patterns of grey stratified, sands and silts. They are moderately acidic throughout and parent alluvium is rich in minerals. Four general soil types occur in the region; of which, Noncalcareous Alluvium predominates. Organic matter content is low and CEC is medium. Soil fertility level, in general, is low to medium.

Location: Narrow belts within and adjoining the channels of the Teesta, Dharla and Dudkumar rivers in Nilphamari, Rangpur, Lalmonirhat, Kurigram and Gaibandha districts.

AEZ-3: Tista Meander Floodplain

This region occupies major part of the Teesta floodplain as well as the floodplains of the Atrai, Little Jamuna, Karatoya, Dharla and Dudkumar rivers.

Most areas have broad floodplain ridges and almost level basins. There is an overall pattern of olive brown, rapidly permeable, loamy soils on the high floodplain ridges, and grey or dark grey, slowly permeable, heavy silt loam or silty clay-loam soils on the lower land and parent materials rich in minerals. Eight general soil types occur in the region; of which, Noncalcareous Grey Floodplain and Noncalcareous Brown Floodplain soils predominate. They are moderately acidic throughout, low in organic matter content on the higher land, but moderate in the power parts. Fertility level, in general, is low to medium. Soils in general have a good moisture holding capacity.

Location: Most of greater Rangpur, eastern parts of Panchagarh and Dinajpur, northern Bogra and parts of Joypurhat, Naogaon and Rajshahi districts.

AEZ-4: Karatoya-Bangali Floodplain

This floodplain apparently comprises a mixture of Tista and Brahmaputra sediments. Most areas have smooth, broad, floodplain ridges and almost level basins.

The soils are grey silt loams and silty clay loams on ridges and grey or dark grey clays in basins. Five general soil types occur in the region; of which, Noncalcareous Grey Floodplain and Noncalcareous Dark Floodplain soils predominate. They are moderately acidic throughout. Organic matter content is low in ridge soils and moderate in basins. General fertility is medium.

Location: Eastern half of Bogra and most of Sirajganj districts.

AEZ-5: Lower Atrai Basin
This region comprises the low lying area between the Barind Tract and the Ganges River Floodplain. Smooth low lying basin land occupies most of the region. Dark grey, heavy, acidic clays predominate.

Seven general soil types occur in the region; but Noncalcareous Dark Grey Floodplain soils cover most of the area. Organic matter and fertility status are moderate.

Location: Naogaon and Natore districts and parts of Rajshahi, Bogra and Sirajgonj districts.

AEZ-6: Lower Punarbhaba Floodplain
This region occupies basins and beels separated by low floodplain ridges. In this area, dark grey, mottled red, very strongly acid, heavy clays occupy both ridge and basin sites. Only one general soil type, Acid Basin Clays, has been identified in the region. Organic matter status is medium to high. General fertility level is medium.

Location: Extreme western part of Naogaon district and extreme northern part of Nawabganj district.

AEZ-7: Active Brahmaputra-Jamuna Floodplain
This region comprises the belt unstable alluvial land along the Brahmaputra-Jamuna rivers where land is constantly being formed and eroded by shifting river channels. It has an irregular relief of broad and narrow ridges and depressions.

The area is occupied by sandy and silty alluvium rich in minerals with slightly alkaline in reaction. Six general soil types occupy the area; of which, only Noncalcareous Alluvium predominates. Organic matter status is low and fertility status low to medium.

Location: Eastern parts of Kurigram, Gaibandha, Bogra, Sirajganj and Pabna districts and western parts of Sherpur, Jamalpur, Tangail and Manikganj districts. Also, minor areas in Dhaka, Munshiganj, Narayanganj and Chandpur districts.

AEZ-8: Young Brahmaputra -Jamuna Floodplain
This region comprises of the area of Brahmaputra sediments. It has a complex relief of broad and narrow ridges, inter-ridge depressions, partially in filled cut-off channels and basins.

This area is occupied by permeable silt loam to silty clay loam soils on the ridges and impermeable clays in the basins which are neutral to slightly acidic in reaction. General soil types include predominantly Grey Floodplain soils. Organic matter content is low in ridges and moderate in basins.

Location: Western parts of Jamalpur, Sherpur and Tangail districts, parts of Manikganj Narayanganj and Gajipur districts and a belt adjoining the old Brahmaputra channel through Mymensingh, Kishoreganj and Narsindi districts.

AEZ-9: Old Brahmaputra Floodplain
This region occupies a large area of Brahmaputra sediments before the river shifted to its present Jamuna channel about 200 years ago. The region has broad ridges and basins.

Soils of the area are predominantly silt loams to silty clay loams on the ridges and clay in the basins. General soil types predominantly include Dark Grey Floodplain soil. Organic matter content is low on the ridges and moderate in the basins; topsoils are moderately acidic but subsoils neutral in reaction. General fertility level is low.

Location: Large areas in Sherpur, Jamalpur, Tangail, Mymensingh, Netrokona, Kishoreganj districts. Also, small areas east of Dhaka and Gazipur districts.

AEZ-10: Active Ganges Floodplain
This region occupies unstable alluvial land within and adjoining Gangesriver. It has irregular relief of broad and narrow ridges and depressions.

The area has complex mixtures of calcareous sandy, silty and clayey alluvium. The general soil types predominantly include Calcareous Alluvium and Calcareous Brown Floodplain soils which are low in organic matter and mildly alkaline in reaction. General fertility level is medium.

Location: The region along the Ganges and lower Meghna River channels from the Indian border in Nawabganj and Rajshahi districts to the mouth of Meghna estuary in Lakshmipur and Barisal districts.

AEZ-11: HighGangesRiver Floodplain
This region includes the western part of the Ganges River Floodplain which is predominantly highland and medium highland. Most areas have a complex relief of broad and narrow ridges and inter-ridge depressions, separated by areas with smooth broad ridges and basins.

There is an overall pattern of olive-brown silt loams and silty clay loams on the upper parts of floodplain ridges and dark grey, mottled brown, mainly clay soils on ridge sites and in basins. Most ridge soils are calcareous throughout. General soil types predominantly include Calcareous Dark Grey Floodplain soils and Calcareous Brown Floodplain soils. Organic matter content in brown ridge soils is low and higher in dark grey soils. Soils are slightly alkaline in reaction. General fertility level is low.

Location: Nawabganj, Rajshahi, Southern Pabna, Kushtia, Meherpur, Jessore, Chuadanga, Jhenaidah, Magura, and northern parts of Satkhira and Khulna districts together with minor areas in Noagaon and Narail districts.

AEZ-12: LowGangesRiver Floodplain
The region comprises the eastern half of the Ganges River Floodplain which is low-lying. The region has a typical meander floodplain landscape of broad ridges and basins.

Soils of the region are silt loams and silty clay loams on the ridges and silty clay loams to heavy clays on lower sites. General soil types predominantly include Calcareous Dark Grey and Calcareous Brown Floodplain soils. Organic matter content is low in ridges and moderate in the basins. Soils are calcareous in nature having neutral to slightly alkaline reaction. General fertility level is medium.

Location : Natore, Pabna, Goalanda, Faridpur, Madaripur, Gopalganj and Sariatpur, eastern part of Kushtia, Magura, and Narail, north eastern parts of Khulna and Bagherhat, northern Barisal, and south-western part of Manikganj.

AEZ-13: Ganges Tidal Floodplain

This region occupies an extensive area of tidal floodplain land in south-west of the country. The greater part of this region has smooth relief.

There is a general pattern of grey, slightly calcareous, heavy soils on river banks and grey to dark grey, noncalcareous, heavy silty clays in the extensive basins. Noncalcareous Grey Floodplain soil is the major component of general soil types. Acid Sulphate soil also occupies significant part of the area where it is extremely acidic during dry season. In general, most of the topsoils are acidic and subsoils are neutral to mildly alkaline. Soils of Sundarban area are strongly alkaline. General fertility level is high with medium to high organic matter content.

Location: All or most of Barisal, Jhalakati, Pirojpur, Patuakhali, Barguna, Bagerhat, Khulna, Satkhira districts including Khulna and Bagerhat Sundarban Reserved Forests.

AEZ-14: Gopalganj-Khulna Beels

The region occupies extensive low-lying areas between the Ganges River Floodplain and the Ganges Tidal Floodplain. Almost level, low-lying basins occupy most of the region with low ridges along rivers and creeks.

Soils of the area are grey and dark grey acidic heavy clays overlying peat or muck at 25-100 cm. Soft peat and muck occupy perennially wet basin centres. General soil types include mainly Peat and Noncalcareous Dark Grey Floodplain soils. Organic matter content is medium to high. Fertility level is medium.

Location: A number of separate basin areas in Madaripur, Gopalgonj, Narail, Jessore, Bagerhat and Khulna districts.

AEZ-15: Arial Beel

This region occupies a low lying basin between the Ganges and Dhaleshwari rivers in the south of formerDhaka district.

The soils of this area are dark grey, acidic heavy clays. Noncalcareous Dark Grey Floodplain soils are the major general soil type. Organic matter content generally exceeds two percent in the top and subsoil. Available moisture holding capacity is inherently low. The general fertility level is medium to high.

Location: Munshigonj and Dhaka districts.

AEZ-16: Middle Meghna River Floodplain

This region occupies abandoned channel of the Brahmaputra River on the border between greater Dhaka and Comilla districts. This region includes islands-former Brahmaputra chars within the Meghna River as well as adjoining parts of the mainland.

Soils of the area are grey, loamy on the ridges and grey to dark grey clayey in the basins. Grey sands to loamy sands with compact silty topsoil, occupying areas of old Brahmaputra char. Dominant general type is Noncalcareous Grey Floodplain soils. Topsoils are strongly acidic and subsoils slightly acidic to slightly alkaline. General fertility level is medium.

Location: The region between the southern part of Sylhet Basin and the confluence of the Meghna river with the Dhaleshwari and Ganges rivers covering parts of several districts, Kishoreganj, Brahmanbaria, Comilla, Chandpur, Narsindi and Narayanganj.

AEZ-17: Lower Meghna River Floodplain

This area occupies transitional area between Middle Meghna River Floodplain and the Young Meghna Estuarine Floodplain. The region has slightly irregular relief with little difference in elevation between the ridges and depressions.

Soils of this area are relatively uniform. Silt loams occupy relatively higher areas and silty clay loams the depressions. Noncalcareous Dark Grey Floodplain and Calcareous Grey Floodplain soils are major components of general type. Topsoils are moderately acidic and subsoils neutral in reaction. General fertility level is medium to high with low to medium organic matter status.

Location: Chandpur, Lakshmipur and Noakhali districts.

AEZ-18: Young Meghna Estuarine Floodplain

This region occupies young alluvial land in and adjoining the Meghna estuary. It is almost level with very low ridges and broad depressions. The major soils are grey to olive, deep, calcareous silt loam and silty clay loams and are stratified either throughout or at shallow depth. Calcareous Alluvium and Noncalcareous Grey Floodplain soils are the dominant general type. The soils in the south become saline in dry season. Top soils and subsoils of the area are mildly alkaline. General fertility is medium but low in organic matter.

Location: Chittagong, Feni, Noakhali, Lakshmipur, Bhola, Barisal, Patuakhali and Barguna districts.

AEZ-19: Old Meghna Estuarine Floodplain

This region occupies a large area, mainly low-lying between south of the Surma-Kusiyara Floodplain and northern edge of the Young Meghna Estuarine Floodplain. It comprises smooth, almost level, floodplain ridges and shallow basins.

Silt loam soils predominate on highlands and silty clay to clay in lowlands. Noncalcareous Dark Grey Floodplain soils are the only general type of the area. Organic matter content of the soils are moderate. Moisture holding capacity is medium. Topsoils are moderately acidic, but subsoils neutral in reaction. General fertility level is medium.

Location: Kishoreganj, Habiganj, Brahmanbaria, Comilla, Chandpur, Feni, Noakhali, Lakshmipur, Narsindi, Narayanganj, Dhaka, Sariatpur, Madaripur, Gopalganj and Barisal districts.

AEZ-20: Eastern Surma-Kusiyara Floodplain

This region occupies the relatively higher parts of Surma-Kusiyara Floodplain formed on sediments of the rivers draining into the Meghna catchments area from the hills. The area is mainly smooth broad ridges and basins.

The soils are grey, heavy silty clay loams on the ridges and clays in the basins. Noncalcareous Grey Floodplain soils are the only general type. Organic matter content of soils is moderate. The reaction of soils ranges from strongly acidic to neutral.

Location: Sylhet, Moulvi Bazar, Sunamganj and Habiganj districts.

AEZ-21: Sylhet Basin
The region occupies the lower western side of Surma-Kusiyara Floodplain. The area is mainly smooth broad basins with narrow ridges of higher land along the rivers.

Soils of the area are grey clays in the wet basins and silty clay loams and clay loam on the higher parts which dry out seasonally. Noncalcareous Grey Floodplain soils and Acid Basin Clays are the major components of the general soil types. The soils have moderate content of organic matter and soil reaction is mainly acidic. Fertility level is medium to high.

Location: Large parts of Sunamganj, Habiganj, Netrokona, Kishoreganj and Brahmanbaria districts.

AEZ-22: Northern and Eastern Piedmont Plains
This is a discontinuous region occurring as a narrow strip of land at the foot of the northern and eastern hills. The area comprises merging alluvial fans which slope gently outward from the foot of the hills into smooth low lying basin.

Grey Piedmont soils and Noncalcareous Grey Floodplain soils are the major general soil types of the area. Soils of the area are loams to clays in texture having slightly acidic to strongly acidic reaction. General fertility level is low to medium.

Location: Sherpur, Netrokona, Sunamganj, Sylhet, Moulvi Bazar, Habiganj, Brahmanbaria and Comilla districts.

AEZ-23: Chittagong Coastal Plain
This region occupies the plain land in greater Chittagong district and the eastern part of Feni district. It is a compound unit of piedmont, river, tidal and estuarine floodplain landscapes.

Grey silt loams and silty clay loam soils are predominant. Acid Sulphate soils which are potentially extremely acidic occur in mangrove tidal floodplains. Noncalcareous Grey Floodplain soils, Noncalcareous Alluvium and Acid Sulphate soils are the major components of general soil types of the area. General fertility level of soils is medium. Organic matter content is low to moderate.

Location: Feni, Chittagong and Cox's Bazar districts.

AEZ-24: St. Martin's Coral Island
This small but distinctive region occupies the whole of St. Martin's Island in the extreme south of the country. The area has very gently undulating old beach ridges and inter-ridge depressions surrounded by sandy beaches.

The soils are developed entirely on old and young coral beach sands. Calcareous Alluvium is the only general soil type of the area. General fertility level is low with poor moisture holding capacity.

Location: St. Martin's, Island

AEZ-25: Level Barind Tract

This region is developed over Madhupur Clay. The landscape is almost level and locally irregular along river channels. The predominant soils have a grey silty paddled topsoil with plough pan which either directly overlies grey heavy little weathered Madhupur Clay or merges with the porous silt loam or silty clay loam subsoils having strongly acid clay at greater depth. Shallow Grey Terrace soils and Deep Grey Terrace soils are the major components of general soil types of the area. The soils are low in available moisture holding capacity and slightly acidic to acidic in reaction. Organic matter status is very low and most of the available nutrients are limiting.

Location: Dinajpur, Gaibandha, Joypurhat, Bogra, Naogaon, Sirajganj and Natore districts.

AEZ-26: High Barind Tract

It includes the western part of Barind Tract where the underlying Madhupur Clay has been uplifted and cut into by deep valleys.

The soils include paddled silt loam to silty clay loam in the topsoils and porous silt loam with mottled plastic clay at varying depth. Deep Grey Terrace soils and GreyValley soils are the major components of general soil types of the area. General fertility status is low having low status of organic matter.

Location: Rajshahi, Nawabganj and Naogaon districts.

AEZ-27: North Eastern Barind Tract

This region occupies several discontinuous areas on the north-eastern margins of Barind Tract. It stands slightly higher than adjoining floodplain land.

The region has silty or loamy topsoil and clay loams to clay subsoils and grades into strongly mottled clay. The Madhupur Clay underlying this region is deeply weathered. Deep Red Brown Terrace soil and Deep Grey Terrace soils are the major components of general soil types of the area. The soils are strongly acidic in reaction. Organic matter of the soils is low. General fertility level is poor.

Location: Dinajpur, Rangpur, Gaibandha, Joypurhat and Bogra districts.

AEZ-28: Madhupur Tract

This region of complex relief and soils developed over the Madhupur Clay. The landscape comprises level upland, closely or broadly dissected terrace associated with either shallow or broad, deep valleys.

Eleven general soil types exist in the area; of which, Deep Red Brown Terrace, Shallow Red Brown Terrace soils and Acid Basin Clays are the major ones. The soils on the terrace are better drained, friable clay loams to clays overlying friable clay substratum at varying depth. Soils in the valleys are dark grey heavy clays. They are strongly acidic in reaction with low status of organic matter, low moisture holding capacity and low fertility level.

Location: Dhaka, Gazipur, Narsindi, Narayanganj, Tangail, Mymensingh and Kishoreganj.

AEZ-29: Northern and Eastern Hills

This region includes the country's hill areas. Relief is complex. Hills have been dissected to different degrees over different rocks. In general, slopes are very steep and few low hills have flat summits.

The major hill soils are yellow-brown to strong brown permeable friable loamy, very strongly acidic and low in moisture holding capacity. However, soil patterns generally are complex due to local differences in sand, silt and clay contents of the underlying sedimentary rocks and in the amount of erosion that has occurred. BrownHill soils are the predominant general soil type of the area. Organic matter content and general fertility level is low.

Location: Mainly in Khagrachhari, Chittagong Hill Tracts, Bandarban, Chittagong, Cox's Bazar, Habiganj and Moulvi Bazar districts, small areas along the northern border of Sherpur, Mymensingh, Sunamganj and Sylhet districts in central and south eastern Sylhet and in the east of Brahmanbaria, Comilla and Feni districts.

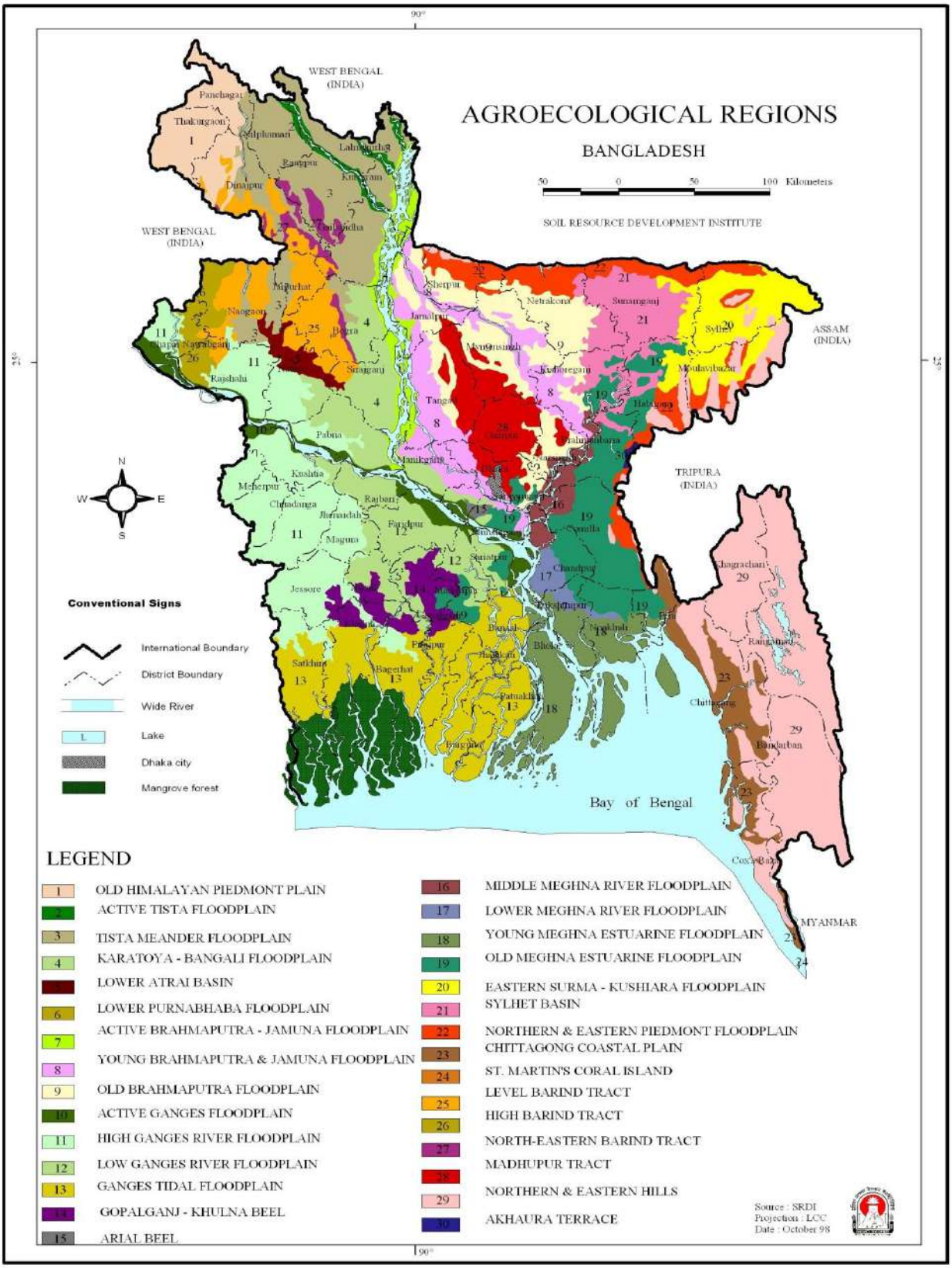
AEZ-30: Akhaura Terrace

This small region lies on the eastern border of Brahmanbaria and southwest corner of Habiganj districts. In appearance, the region resembles Madhupur Tract with level upland dissected by mainly deep broad valleys.

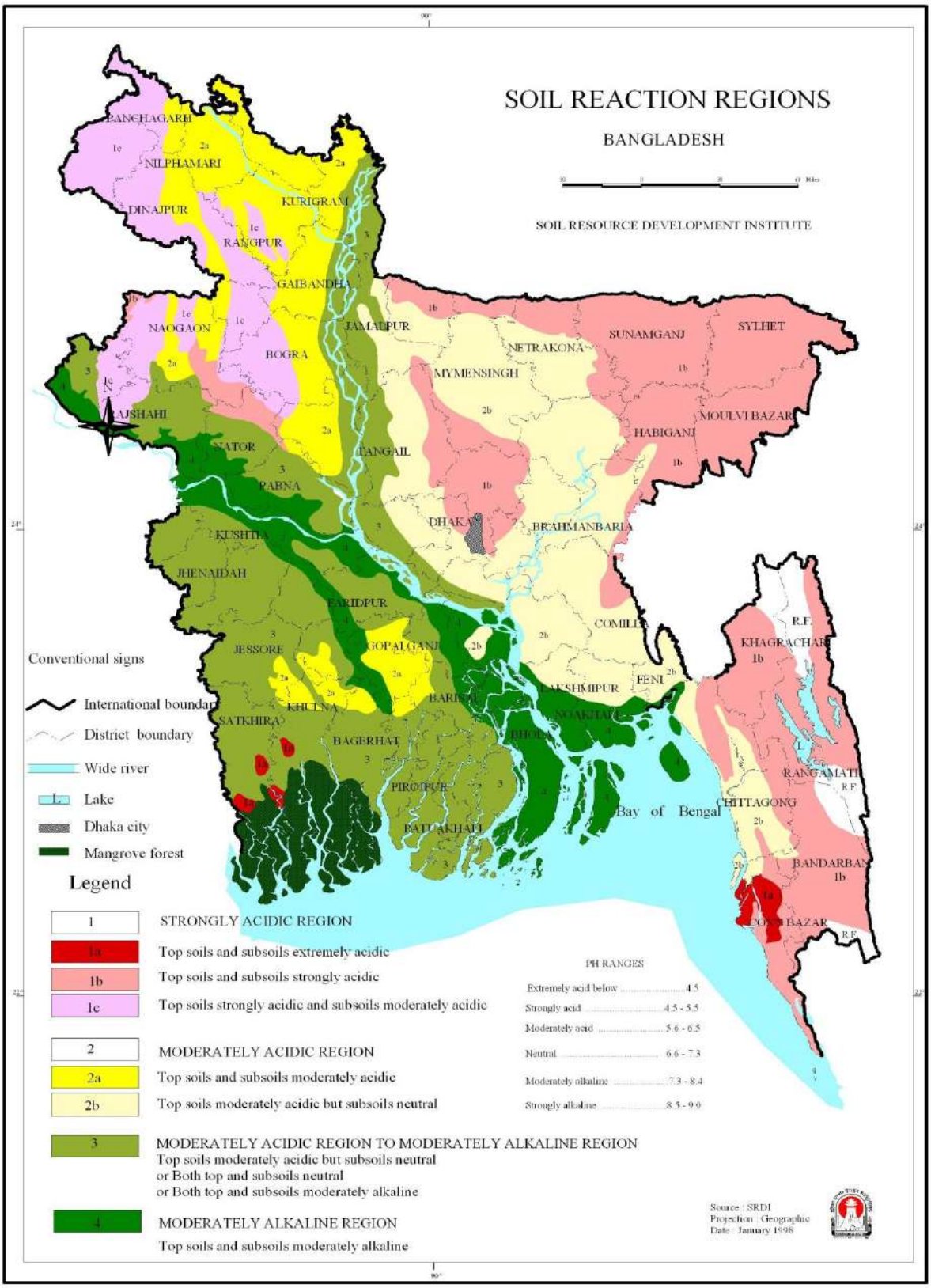
The main soils on the upland have strong brown clay which grades into red mottled clay substratum. The valley soils range from silty clay loams to clays. Deep Red Brown Terrace soil, Grey Piedmont soils and Acid Basin Clays are the major components of general soil types of the area. The general fertility including organic matter status is low. The soils are strongly acidic in reaction.

Location: Brahmanbaria district and minor area in Habiganj district.

All the physiographic units and agro-ecological regions are displayed separately on two maps (Map-1 and Map-2).


AGROECOLOGICAL REGIONS
BANGLADESH
50 0 50 100 Kilometers
SOIL RESOURCE DEVELOPMENT INSTITUTE
WEST BENGAL (INDIA)
ASSAM (INDIA)
TRIPURA (INDIA)
MYANMAR
Bay of Bengal
Conventional Signs
International Boundary
District Boundary
Wide River
Lake
Dhaka city
Mangrove forest
LEGEND
1 OLD HIMALAYAN PIEDMONT PLAIN
2 ACTIVE TISTA FLOODPLAIN
3 TISTA MEANDER FLOODPLAIN
4 KARATOYA - BANGALI FLOODPLAIN
5 LOWER ATRAI BASIN
6 LOWER PURNABHABA FLOODPLAIN
7 ACTIVE BRAHMAPUTRA - JAMUNA FLOODPLAIN
8 YOUNG BRAHMAPUTRA & JAMUNA FLOODPLAIN
9 OLD BRAHMAPUTRA FLOODPLAIN
10 ACTIVE GANGES FLOODPLAIN
11 HIGH GANGES RIVER FLOODPLAIN
12 LOW GANGES RIVER FLOODPLAIN
13 GANGES TIDAL FLOODPLAIN
14 GOPALGANJ - KHULNA BEEL
15 ARIAL BEEL
16 MIDDLE MEGHNA RIVER FLOODPLAIN
17 LOWER MEGHNA RIVER FLOODPLAIN
18 YOUNG MEGHNA ESTUARINE FLOODPLAIN
19 OLD MEGHNA ESTUARINE FLOODPLAIN
20 EASTERN SURMA - KUSHIARA FLOODPLAIN SYLHET BASIN
21 NORTHERN & EASTERN PIEDMONT FLOODPLAIN
22 CHITTAGONG COASTAL PLAIN
23 ST. MARTIN'S CORAL ISLAND
24 LEVEL BARIND TRACT
25 HIGH BARIND TRACT
26 NORTH-EASTERN BARIND TRACT
27 MADHUPUR TRACT
28 NORTHERN & EASTERN HILLS
29 AKHAURA TERRACE
Source : SRDI
Projection : LCC
Date : October 98

Map-1


SOIL REACTION REGIONS
BANGLADESH
SOIL RESOURCE DEVELOPMENT INSTITUTE
PANCHAGARH
NILPHAMARI
DINAJPUR
KURIGRAM
RANGPUR
GAIBANDHA
NAOGAON
BOGRA
JAMALPUR
SUNAMGANJ
SYLHET
NETRAKONA
MYMENSINGH
RAJSHAHI
MOULVI BAZAR
HABIGANJ
SATOR
TANGAIL
RABNA
DHAKA
BRAHMANBARIA
KUSHTIA
JHENAIDAH
FARIDPUR
COMILLA
JESSORE
GOPALGANJ
KHAGRACHARI
FENI
LAKSHMIPUR
KHULNA
BARISAL
SATKHIRA
NOAKHALI
BAGERHAT
RANGAMATI
CHITTAGONG
PIROJPUR
Bay of Bengal
PATUAKHALI
BANDARBAN
COX'S BAZAR
Conventional signs
International boundary
District boundary
Wide river
Lake
Dhaka city
Mangrove forest
Legend
1 STRONGLY ACIDIC REGION
1a Top soils and subsoils extremely acidic
1b Top soils and subsoils strongly acidic
1c Top soils strongly acidic and subsoils moderately acidic
2 MODERATELY ACIDIC REGION
2a Top soils and subsoils moderately acidic
2b Top soils moderately acidic but subsoils neutral
3 MODERATELY ACIDIC REGION TO MODERATELY ALKALINE REGION
Top soils moderately acidic but subsoils neutral
or Both top and subsoils neutral
or Both top and subsoils moderately alkaline
4 MODERATELY ALKALINE REGION
Top soils and subsoils moderately alkaline
PH RANGES
Extremely acid below 4.5
Strongly acid 4.5 - 5.5
Moderately acid 5.6 - 6.5
Neutral 6.6 - 7.3
Moderately alkaline 7.3 - 8.4
Strongly alkaline 8.5 - 9.0
Source : SRDI
Projection : Geographic
Date : January 1998

Map-2

1.5 Land Levels In Relation to Flooding

The distribution of land levels in relation to seasonal flooding is shown in Map-3. The information has been provided in terms of depth of flooding phases. The terms used have the following meanings:

Highland (H)	: Land which is above normal flood-level.
Medium Highland (MH)	: Land which is normally flooded about 90 cm deep during the flood season.
Medium Lowland (ML)	: Land which is normally flooded between 90 cm. and 180 cm. deep during the flood season.
Lowland (L)	: Land which is normally flooded between 180 and 300 cm deep during the flood season.
Very Lowland (VL) :	Land which is normally flooded deeper than 300 cm during the flood season.

An additional class, Bottomland, is recognized for depression sites in any land level class which remains wet throughout the year. The Medium Highland land type has been divided into two subclasses: MH-1, normally flooded upto about 30 cm deep; and MH-2, normally flooded between 30 -90 cm deep. Neither these subclasses nor Bottomland are shown separately in the map.

It is important to realize that the depth limits between the depth of flooding classes are not as rigid as the definitions might suggest. Flood-levels in an area may, in fact, vary by as much as a metre or more over different years. They may also reach their peak levels for only a few days at a time during a particular year. They may also reach their peak levels for only a few days at a time during a particular year. What these classes indicate is the level of flooding which farmers expect when they decide which crops to grow in the kharif season on their different kinds of land, based on their long experience of cultivation on particular sites:

- Highland may be suitable for kharif or perennial dry land crops if the soils are permeable. Impermeable soils - or soils which can be made impermeable by puddling - may be suitable for transplanted Aus and/or Aman paddy if bunds are made to retain rainwater on fields.

- Medium Highland is suitable for crops which can tolerate shallow flooding, such as broadcast or transplanted Aus paddy, jute and transplanted Aman paddy. Early kharif dry land crops which mature before flooding starts can be grown on permeable soils, and late kharif and early rabi dry land crops on soils which drain in September-October.

- Medium Lowland is flooded too deeply for transplanted Aus or transplanted Aman paddy to be grown safely. Mixed broadcast Aus and deepwater Aman is a common practice; or long Aman seedlings may be transplanted if floodwater recedes early enough. Dry land rabi crops are widely grown on soils which drain in October or November.

- Lowland is flooded too deeply for broadcast Aus or transplanted Aman to be grown. Deepwater Aman is typically grown on such land (although the cultivation of irrigated Boro paddy on such land in the dry season now precludes the cultivation of deepwater Aman over considerable areas of lowland). Dry land rabi crops can only be grown if floodwater recedes before December.

- Very Lowland generally is too deeply flooded for even deepwater Aman to be grown (this is not necessary because of the depth of flooding but because of such associated characteristics as early flooding, rapid flooding or wave action on large open bodies of water, as in the Sylhet Basin). Where cultivated, very Lowland is generally used for irrigated Boro paddy, either HYV or local varieties.

- Bottomland stays too wet for paddy to be sown broadcast. The traditional crop on such land is local Boro paddy, either not irrigated or irrigated by traditional low-lift irrigation devices. In a few other areas where flooding normally does not exceed about 1.5 m deep very long Aman paddy seedlings are transplanted early in the monsoon season.

The above flood-depth classes are shown on the map as subdivisions of the soil units. As with the soil units themselves, it must be realized that the information given on flooding characteristics is highly generalized. It is to be noted that several flood-depth classes can occur within a single village in floodplain physiographic units. But the map shows either the most extensive class (or classes) occurring in the soil unit (e.g., Highland or Highland and Medium Highland) or the range of classes which cover important areas within the unit if more than two classes occur (e.g., Medium Lowland to Very Lowland). Where two classes are named, the more extensive one is named first. Where a range of classes is indicated, they are referred in sequence from highest to lowest position in the landscape, regardless of their relative extents.

Inundation Land Types

1. Mainly Highland
2. Highland to Medium Lowland
3. Mainly Medium Highland
4. Medium Highland and Highland
5. Medium Highland and Medium Lowland
6. Medium Highland to Lowland
7. Medium Lowland and Lowland
8. Lowland and Very Lowland